SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project: Hospital Management System

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SECTION K

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Hospital Management System. It defines the functional and non-functional requirements, external interfaces, security requirements, and verification criteria of the system. The document is intended to provide a clear reference for developers, testers, hospital staff, administrators, and project evaluators.

1.2 Scope

The Hospital Management System provides computerized management of hospital operations including patient registration, doctor management, appointment scheduling, medical records, prescriptions, billing, payments, and notifications. The system allows authorized hospital staff and doctors to access and manage information according to their roles.

The system does not include the internal implementation of external payment gateways, external laboratory systems, or external insurance systems beyond their required interfaces.

1.3 Audience

The intended audience includes software developers, QA engineers, doctors, nurses, receptionists, pharmacists, hospital administrators, system administrators, and project evaluators.

1.4 Definitions

HMS – Hospital Management System

Patient – A person receiving medical services from the hospital.

Doctor – A medical professional who provides consultation, diagnosis, and treatment.

Administrator – An authorized user responsible for managing hospital users, departments, and system configuration.

Receptionist – A hospital staff member responsible for patient registration, appointment scheduling, and billing-related activities.

Pharmacist – A hospital staff member responsible for managing prescriptions and dispensing medicines.

Appointment – A scheduled consultation between a patient and a doctor.

EMR – Electronic Medical Record.

UI – User Interface.

API – Application Programming Interface.

2. Overall Description

2.1 Product Perspective

The Hospital Management System is a centralized software system used to manage hospital activities and patient information. It provides interfaces for patients, doctors, receptionists, pharmacists, and administrators. The system stores patient records, appointments, prescriptions, billing information, and other operational data in a centralized database.

2.2 Major Product Functions

The major functions of the Hospital Management System include user registration and authentication, patient registration and profile management, doctor profile and availability management, appointment scheduling and cancellation, patient medical record management, diagnosis and treatment recording, prescription management, billing and payment management, appointment notifications and reminders, and report generation.

2.3 User Roles and Characteristics

Patient: Can register, manage personal information, book appointments, view medical records, prescriptions, bills, and receive notifications.

Doctor: Can view scheduled appointments, access authorized patient medical records, record diagnosis and treatment details, and create prescriptions.

Receptionist: Can register patients, schedule appointments, manage basic patient information, and generate bills.

Pharmacist: Can view prescriptions, manage medicine availability, and update medicine dispensing records.

Administrator: Can manage users, doctors, hospital departments, system configuration, and generate reports.

System Administrator: Responsible for system maintenance, security, backups, monitoring, and technical administration.

2.4 Operating Environment

The system shall operate on hospital computers, laptops, tablets, or other supported devices using a modern web browser or supported application interface. The system shall use a centralized database server and a secure hospital network or Internet connection. The system shall support the operating systems, database management systems, and software platforms required for deployment.

2.5 Constraints

The system shall restrict access to patient medical information to authorized users. The system shall implement role-based access control. Personal and medical data shall be protected during storage and transmission. The system shall comply with applicable healthcare data protection and privacy requirements. The system shall depend on the availability of the hospital network and database server. External payment, laboratory, insurance, SMS, or email services may require third-party APIs.

3. External Interface Requirements

3.1 User Interfaces

The Hospital Management System shall provide a clear and user-friendly interface for patients, doctors, receptionists, pharmacists, and administrators. Each user shall be provided with a role-specific dashboard and access to functions permitted for that role. Forms shall provide validation messages for incorrect, incomplete, or invalid data. The interface shall allow users to easily navigate between patient records, appointments, prescriptions, billing, and other relevant functions.

3.2 Hardware Interfaces

The system shall support desktop computers, laptops, tablets, and other supported devices used by hospital staff. The system may interface with printers for printing bills, prescriptions, and reports. Barcode or QR-code scanners may be supported for patient identification, medicine management, or other applicable hospital operations.

3.3 Software Interfaces

The system shall interface with a centralized database for storing and retrieving hospital information. It may integrate with external payment gateway APIs for processing payments and SMS or email services for sending notifications. The system may also integrate with laboratory, pharmacy, or insurance systems through appropriate APIs where required.

3.4 Communications

HTTPS using TLS 1.2 or higher shall be used for secure transmission of sensitive information between clients and the Hospital Management System.

4. System Features

4.1 User Authentication

Description: Authenticate users and enforce role-based access to protected system functions.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-001	The system shall authenticate users using a valid username or email and password.	Functional	High	Security / Hospital Administrator	Valid credentials create an authenticated session. **TC-Auth-01**	Requires user account store and authentication service.
HMS-F-002	The system shall provide role-based access for patients, doctors, receptionists, pharmacists, and administrators.	Functional	High	Security / Hospital Administrator	Users can access only functions permitted for their assigned role. **TC-Auth-02**	RBAC configuration required.
HMS-F-003	The system shall allow authenticated users to securely log out and terminate their active session.	Functional	High	Security / All Users	After logout, protected functions cannot be accessed without authentication. **TC-Auth-03**	Requires session management.

4.2 Patient Management

Description: Manage patient registration, profiles, and authorized access to patient information.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-004	The system shall allow patients or authorized receptionists to register a new patient and create a patient profile containing required personal and contact information.	Functional	High	Patient / Receptionist	A valid patient registration creates a unique patient profile. **TC-Pat-01**	Requires patient database.
HMS-F-005	The system shall allow authorized users to view and update permitted	Functional	High	Receptionist / Administrator / Patient	Authorized users can update permitted	Requires role-based permissions.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	patient demographic and contact information.				information and changes are saved correctly. **TC-Pat-02**	
HMS-F-006	The system shall allow authorized doctors and hospital staff to search for and access patient records according to their assigned permissions.	Functional	High	Doctor / Hospital Staff	Authorized users can retrieve permitted patient records while unauthorized users are denied access. **TC-Pat-03**	Requires patient database and RBAC.

4.3 Doctor Management

Description: Manage doctor profiles, availability, and scheduled appointments.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-007	The system shall allow administrators to create and manage doctor profiles, including specialization, department, and available consultation time slots.	Functional	High	Hospital Administrator	Administrator can create and update doctor information and availability. **TC-Doc-01**	Requires doctor and department data.
HMS-F-008	The system shall allow doctors to view their scheduled appointments and relevant patient information.	Functional	High	Doctor	Doctor can view upcoming appointments and permitted patient details. **TC-Doc-02**	Depends on appointment and patient modules.

4.4 Appointment Management

Description: Schedule, validate, cancel, reschedule, and notify users about appointments.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-009	The system shall allow patients or authorized receptionists to schedule an appointment with an available doctor by selecting a suitable date and time slot.	Functional	High	Patient / Receptionist / Doctor	A valid available time slot can be booked successfully. **TC-Apt-01**	Requires doctor availability and patient records.
HMS-F-010	The system shall validate doctor availability and prevent two appointments from being scheduled for the same doctor and time slot.	Functional	High	Hospital Administrator / Doctor / Receptionist	The system rejects an appointment when the selected doctor and time slot are already occupied. **TC-Apt-02**	Depends on appointment database and scheduling logic.
HMS-F-011	The system shall allow authorized users to cancel or reschedule an existing appointment according to hospital policies.	Functional	High	Patient / Receptionist / Doctor	Authorized users can cancel or reschedule appointments and the updated status is stored correctly. **TC-Apt-03**	Requires appointment status management.
HMS-F-012	The system shall send appointment confirmation and reminder notifications to patients.	Functional	Medium	Patient / Hospital Administration	Confirmation and reminder notifications are generated for scheduled appointments. **TC-Apt-04**	Requires notification service such as email or SMS.

4.5 Medical Records

Description: Maintain patient medical records, diagnosis, treatment details, and authorized access to historical records.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-013	The system shall allow authorized doctors to create and update patient medical records after a consultation.	Functional	High	Doctor / Hospital Administration	Doctor can create or update a patient's medical record and the information is stored successfully. **TC-Med-01**	Requires patient record database and authorization.
HMS-F-014	The system shall allow authorized doctors to record diagnosis, treatment details, consultation notes, and other relevant clinical information.	Functional	High	Doctor	Diagnosis and treatment information entered by an authorized doctor is stored against the correct patient record. **TC-Med-02**	Requires medical record module.
HMS-F-015	The system shall allow authorized doctors and permitted hospital staff to view a patient's previous medical records according to their access permissions.	Functional	High	Doctor / Hospital Staff	Authorized users can view previous records while unauthorized users are denied access. **TC-Med-03**	Requires RBAC and medical record database.

4.6 Prescription Management

Description: Create electronic prescriptions and manage medicine dispensing

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-	The system shall	Functional	High	Doctor	Doctor can	Requires patient

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
016	allow doctors to create electronic prescriptions containing medicines, dosage, frequency, duration, and instructions.				create and save a complete prescription for a patient. **TC-Pres-01**	and medicine information.
HMS-F-017	The system shall allow pharmacists to view prescriptions authorized for dispensing and record the medicines that have been dispensed.	Functional	High	Pharmacist	Pharmacist can view an authorized prescription and record the medicines dispensed. **TC-Pres-02**	Requires prescription and medicine inventory data.

4.7 Billing and Payments

Description: Generate hospital bills and record payments against outstanding charges.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-018	The system shall allow authorized receptionists or billing staff to generate bills for consultations, medical services, laboratory tests, medicines, and other applicable hospital charges.	Functional	High	Receptionist / Billing Staff	A bill is generated with the correct patient details, services, charges, and total amount. **TC-Bill-01**	Requires patient, service, and billing data.
HMS-F-019	The system shall allow authorized staff to record payments against generated bills and update the payment status.	Functional	High	Receptionist / Billing Staff	A valid payment updates the corresponding bill status correctly. **TC-Bill-02**	Requires billing module and payment records.

4.8 Reports and Notifications

Description: Provide administrators with operational reports for monitoring and management.

Req ID	Requirement	Type	Priority	Source/Stake holder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-020	The system shall allow administrators to generate reports related to patients, appointments, billing, doctors, and other hospital operations.	Functional	Medium	Hospital Administrator	Administrator can select a report type and generate a report containing the relevant hospital data. **TC-Rep-01**	Requires access to relevant hospital databases.

5. Non-functional Requirements (Detailed)

NFRs below are measurable and tied to test plans. IDs follow **HMS-NF-###**.

5.1 Non-functional Requirements

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
HMS-NF-001	The system shall provide a response time of 3 seconds or less for at least 90% of transactions under normal operating load.	Performance	High	At least 90% of transactions complete within 3 seconds during production-like load testing. **TC-Perf-01**
HMS-NF-002	The system shall provide at least 99.5% monthly availability, excluding scheduled maintenance.	Reliability	High	Monthly system uptime shall be at least 99.5%. **TC-Rel-01**
HMS-NF-003	The system shall protect sensitive patient and user information and shall encrypt sensitive information during transmission.	Security / Compliance	High	Secure encrypted connections are used and unauthorized access attempts are rejected. **TC-Sec-01**
HMS-NF-004	The system shall maintain data consistency and prevent loss or corruption of patient, appointment, medical, prescription, and billing records.	Data Integrity / Reliability	High	Records remain consistent after normal operations, failed transactions, and system recovery. **TC-Data-01**
HMS-NF-005	The system shall provide a clear and consistent user interface that allows common hospital tasks to be completed with minimal training.	Usability / Accessibility	Medium	Representative users can complete common tasks with minimal training and without critical usability issues. **TC-UX-01**

5.2 Security

5.2.1 Security Objectives

Objective ID	Security Objective	Priority	Test Case
SO-001	**Confidentiality:** The system shall protect patient personal and medical information from unauthorized access or disclosure.	High	TC-SEC-01
SO-002	**Integrity:** The system shall prevent unauthorized modification or deletion of patient, medical, appointment, prescription, and billing information.	High	TC-SEC-02

Objective ID	Security Objective	Priority	Test Case
SO-003	**Availability:** The system shall ensure that authorized hospital users can access required system services and records when needed.	High	TC-SEC-03

5.2.2 Security Requirements

Req ID	Security Requirement	Category	Priority	Acceptance Criteria / Test Case Ref
HMS-SR-001	The system shall enforce role-based access control for patients, doctors, receptionists, pharmacists, administrators, and system administrators.	Security	High	Users can access only functions permitted for their assigned role. **TC-Sec-04**
HMS-SR-002	The system shall use TLS 1.2 or higher for transmitting sensitive information between clients and servers.	Security	High	Sensitive information is transmitted only through an approved secure connection. **TC-Sec-05**
HMS-SR-003	The system shall store user passwords using strong one-way hashing and shall never store passwords in plain text.	Security	High	Password storage is verified to use secure one-way hashing. **TC-Sec-06**
HMS-SR-004	The system shall maintain audit logs for authentication events, patient record modifications, billing changes, prescription changes, and administrative actions.	Security	High	Required security-sensitive actions are recorded in audit logs. **TC-Sec-07**
HMS-SR-005	The system shall automatically terminate or lock inactive user sessions after a configured period of inactivity.	Security	High	An inactive session is automatically terminated or locked after the configured timeout. **TC-Sec-08**

5.3 Summary

The Hospital Management System defines functional, non-functional, and security requirements to ensure that hospital operations are managed efficiently and securely. The requirements cover user authentication, patient and doctor management, appointment scheduling, medical records, prescriptions, billing, payments, reports, and notifications.

The non-functional requirements specify measurable expectations for system performance, reliability, security, data integrity, and usability. The security objectives and security requirements ensure confidentiality, integrity, and availability of hospital and patient information through role-based access control, secure communication, password protection, audit logging, and session management.

These requirements provide the basis for system development, testing, security verification, and acceptance of the Hospital Management System.

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

The Hospital Management System shall provide reliable, secure, usable, maintainable, and performant services to its authorized users.

The system shall satisfy the following quality attributes:

- **Performance:** The system should respond to user requests within the specified response-time requirements under normal operating conditions.
- **Reliability:** The system should remain available and operate consistently with minimal downtime.
- **Security:** Patient, medical, billing, prescription, and user information shall be protected from unauthorized access and modification.
- **Data Integrity:** The system shall maintain accurate and consistent hospital records and prevent data loss or corruption.
- **Usability:** The system shall provide a clear and consistent interface that allows hospital staff and patients to perform common tasks with minimal training.
- **Maintainability:** The system should be structured so that authorized technical staff can maintain, update, and troubleshoot the system efficiently.

6.2 Acceptance Criteria

The Hospital Management System shall be considered acceptable when:

1. All specified functional requirements have been implemented and verified.
2. All specified non-functional requirements have been tested against their defined measurements.
3. All specified security requirements have passed their corresponding security tests.
4. Patient, appointment, medical, prescription, and billing data are stored and retrieved correctly.
5. Role-based access control correctly restricts users to their permitted functions.
6. No critical defects remain unresolved at the time of acceptance.
7. Each requirement can be traced to a corresponding test case through the Requirements Traceability Matrix (RTM).

6.3 Acceptance Test Suites

Test Suite ID	Test Suite	Purpose
TC-AUTH	Authentication and Authorization Tests	Verify user login, logout, authentication, and role-based access control.
TC-PAT	Patient Management Tests	Verify patient registration, profile management, and patient record access.
TC-DOC	Doctor Management Tests	Verify doctor profile, availability, and appointment-related functions.
TC-APT	Appointment Management Tests	Verify appointment scheduling, availability validation, cancellation, rescheduling, and notifications.
TC-MED	Medical Record Tests	Verify creation, modification, and authorized viewing of medical records.
TC-PRES	Prescription Tests	Verify prescription creation and medicine dispensing functions.
TC-BILL	Billing and Payment Tests	Verify bill generation, payment recording, and payment status updates.
TC-PERF	Performance Tests	Verify system response time and performance under normal load.
TC-SEC	Security Tests	Verify confidentiality, integrity, access control, secure communication, password protection, audit logging, and session security.
TC-UX	Usability Tests	Verify that representative users can perform common tasks with minimal training.

6.4 Acceptance Test Exit Criteria

The acceptance testing phase shall be considered complete when all specified requirements have passed their corresponding test cases, no critical defects remain unresolved, and the system satisfies the defined functional, non-functional, and security requirements.

7. UML Use-Case Diagrams

Diagram 7.1 — Patient and Doctor



Diagram 7.2 — Receptionist, Pharmacist and Administrator


Hospital Management System
Register Patient
Schedule Appointment
<<include>>
<<include>>
Select Date and Time
Check Doctor Availability
Cancel / Reschedule Appointment
Receptionist
<<extend>>
Provide Cancellation Reason
Generate Bill
Generate Reports
Administrator
Record Payment
View Prescription
Pharmacist
Manage Users, Doctors and Departments
Record Medicine Dispensing

8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix maps each requirement to its corresponding system module and verification test case.

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
HMS-F-001	User authentication	4.1	Authentication	TC-Auth-01	N	Depends on user account database.
HMS-F-002	Role-based access	4.1	Authentication	TC-Auth-02	N	Depends on role-based access configuration.
HMS-F-003	Secure logout	4.1	Authentication	TC-Auth-03	N	Requires secure session management.
HMS-F-004	Patient registration	4.2	Patient Management	TC-Pat-01	N	Requires patient database.
HMS-F-005	Patient profile management	4.2	Patient Management	TC-Pat-02	N	Access depends on assigned user role.
HMS-F-006	Patient record access	4.2	Patient Management	TC-Pat-03	N	Requires authorization and patient record access control.
HMS-F-007	Doctor profile management	4.3	Doctor Management	TC-Doc-01	N	Requires doctor and department information.
HMS-F-008	Doctor appointment viewing	4.3	Doctor Management	TC-Doc-02	N	Depends on appointment and patient modules.
HMS-F-009	Appointment scheduling	4.4	Appointment Management	TC-Apt-01	N	Requires doctor availability and patient information.
HMS-F-010	Appointment availability validation	4.4	Appointment Management	TC-Apt-02	N	Prevents conflicting appointment time slots.
HMS-F-011	Appointment cancellation/rescheduling	4.4	Appointment Management	TC-Apt-03	N	Subject to hospital appointment policies.
HMS-F-012	Appointment notifications	4.4	Appointment Management	TC-Apt-04	N	Requires email or SMS notification service.
HMS-F-013	Medical record management	4.5	Medical Records	TC-Med-01	N	Requires authorized doctor access.
HMS-F-014	Diagnosis and treatment recording	4.5	Medical Records	TC-Med-02	N	Requires medical record storage.

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
HMS-F-015	Medical history access	4.5	Medical Records	TC-Med-03	N	Access depends on user permissions.
HMS-F-016	Electronic prescriptions	4.6	Prescription Management	TC-Pres-01	N	Requires patient and medicine information.
HMS-F-017	Medicine dispensing	4.6	Prescription Management	TC-Pres-02	N	Requires prescription and medicine inventory data.
HMS-F-018	Bill generation	4.7	Billing and Payments	TC-Bill-01	N	Requires patient, service, and billing information.
HMS-F-019	Payment recording	4.7	Billing and Payments	TC-Bill-02	N	Requires billing and payment records.
HMS-F-020	Hospital reports	4.8	Reports and Notifications	TC-Rep-01	N	Requires access to relevant hospital data.
HMS-NF-001	Response time	5.1	System	TC-Perf-01	N	Verify under normal operating load.
HMS-NF-002	System availability	5.1	System	TC-Rel-01	N	Excludes scheduled maintenance periods.
HMS-NF-003	Data security	5.1	Security	TC-Sec-01	N	Sensitive data must use secure transmission.
HMS-NF-004	Data integrity	5.1	System	TC-Data-01	N	Verify during normal operation and recovery testing.
HMS-NF-005	Usability	5.1	User Interface	TC-UX-01	N	Evaluate using representative hospital users.
SO-001	Confidentiality	5.2.1	Security	TC-SEC-01	N	Applies to patient and medical information.
SO-002	Integrity	5.2.1	Security	TC-SEC-02	N	Applies to critical hospital records.
SO-003	Availability	5.2.1	Security	TC-SEC-03	N	Depends on system availability and recovery mechanisms.
HMS-SR-001	Role-based security	5.2.2	Security	TC-Sec-04	N	Applies to all defined user roles.
HMS-SR-002	Secure communication	5.2.2	Security	TC-Sec-05	N	Applies to client-server communication.
HMS-SR-003	Password protection	5.2.2	Security	TC-Sec-06	N	Passwords must never be stored in

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
						plain text.
HMS-SR-004	Audit logging	5.2.2	Security	TC-Sec-07	N	Security-sensitive actions must be recorded.
HMS-SR-005	Session timeout	5.2.2	Security	TC-Sec-08	N	Timeout duration should be configurable.

Status Legend:

- **N** – Not yet tested/implemented
- **P** – Partially tested/implemented
- **A** – Accepted/approved